

D Proof of Lemma 5.2

Lemma 5.2. *When updating an MLP layer of an LM using backpropagation only with the VJPs of the i -th token, δ_i , and rerunning the layer with the same inputs x_i from the forward pass of the prompt we used for the editing, the following occurs: (i) The inputs, x_i , are added to or subtracted from the neurons of FF_1 , thereby adjusting how much the activations of each corresponding neuron in FF_2 increase or decrease. (ii) The VJPs δ_i are subtracted from the neurons of FF_2 , amplifying in FF_2 ’s output the presence of the VJPs after they are multiplied with negative coefficients.*

Proof. In Section 4.2, we revealed that FF_1 weights are updated by injections (adding or subtracting vectors) of its x_i . This update is done according to the coefficients of its δ_i , after multiplying them with the learning rate. If we rerun the same layer with the same input after the update, we obtain the following output per each neuron j :

$$\begin{aligned} x_i \cdot (FF_1[j] + \eta \cdot x_i^\top \cdot \delta_i[j]) &= \\ x_i \cdot FF_1[j] + x_i \cdot \eta \cdot x_i^\top \cdot \delta_i[j] &= \\ x_i \cdot FF_1[j] + \|x_i\|_2^2 \cdot \eta \cdot \delta_i[j] &\in \mathbb{R} \end{aligned} \quad (17)$$

$x_i \cdot FF_1[j]$ is the pre-edit output of this neuron. The second component, $\|x_i\|_2^2 \cdot \eta \cdot \delta_i[j]$, is derived from the update and can be positive or negative, hence it controls the increment or decrement of this output compared to the pre-edit one. In models with monotonic (or semi-monotonic) activation functions, such as ReLU (GeLU is positive monotonic only from around -0.75), the activation of the corresponding neuron in FF_2 will be changed directly by this addition to that output.

In Section 4.2, we show how FF_2 ’s δ_i form the gradient matrix. Consider the result of updating only FF_2 and rerunning the same layer:

$$\begin{aligned} x_i[j] \cdot (FF_2[j] + \eta \cdot \delta_i \cdot x_i[j]) &= \\ x_i[j] \cdot FF_2[j] + \eta \cdot (x_i[j])^2 \cdot \delta_i &\in \mathbb{R}^d \end{aligned} \quad (18)$$

$x_i[j] \cdot FF_2[j]$ represents the original output of this neuron. Since $(x_i[j])^2$ is always non-negative and η is negative, the update is a subtraction of δ_i from each neuron.

□

Figure 10 illustrates “imprint and shift” mechanism.

E Additional Logit Lens of Gradients Examples

The work that presented Logit Lens (LL) (nostalgebraist, 2020) established this method by constructing tables of the projected results of different prompts, illustrating the tokens each individual forward pass represents at each layer of the model. In this section, we present a similar approach, focusing on the backward pass rather than the forward pass. In the following figures, we provide examples of the LL projections of gradients when they are interpreted as combinations of the intermediate inputs x_i or VJPs δ_i according to Section 4.2.

In addition to illustrating the information stored in the gradient matrices, the following tables also describe the information moving through LMs during the forward and backward pass. Our interpretation of FF_1 using x_i reflects the gradual buildup in the prediction of the forward pass, as x_i represents the intermediate inputs for each MLP layer. However, FF_2 ’s δ_i , VJPs, serve as the backpropagation counterpart to the forward pass x_i . We can conceptualize them as the input of the MLP when executing the backward pass, or as the error propagated from later layers to earlier ones. In conclusion, Figure 11, 12, 13, 14, 15 depict the information stored within the gradients, simultaneously also comparing the information revealed by LL from the forward pass (x_i , known from prior studies) with that from the backward pass (δ_i), which is part of our innovative contribution.

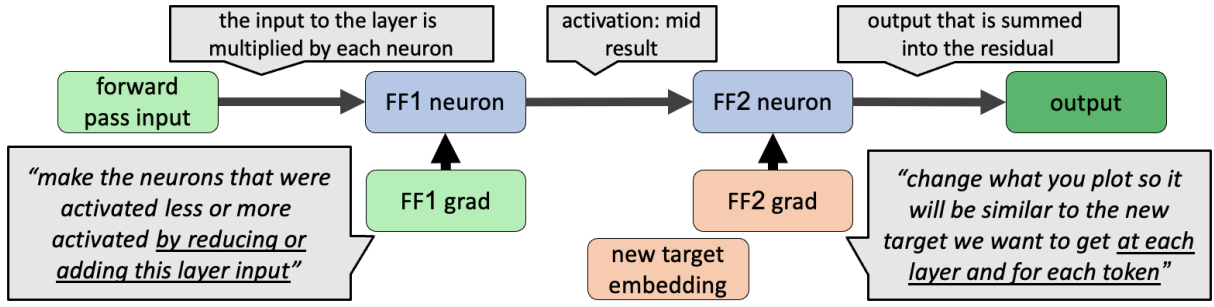


Figure 10: The **Imprint and Shift** mechanism of backpropagation. “grad” represents a single neuron from a gradient matrix. The color of FF_1 grad is the same as the forward-pass input, while FF_2 is the same as the new target embedding, suggesting that they are similar to each other.

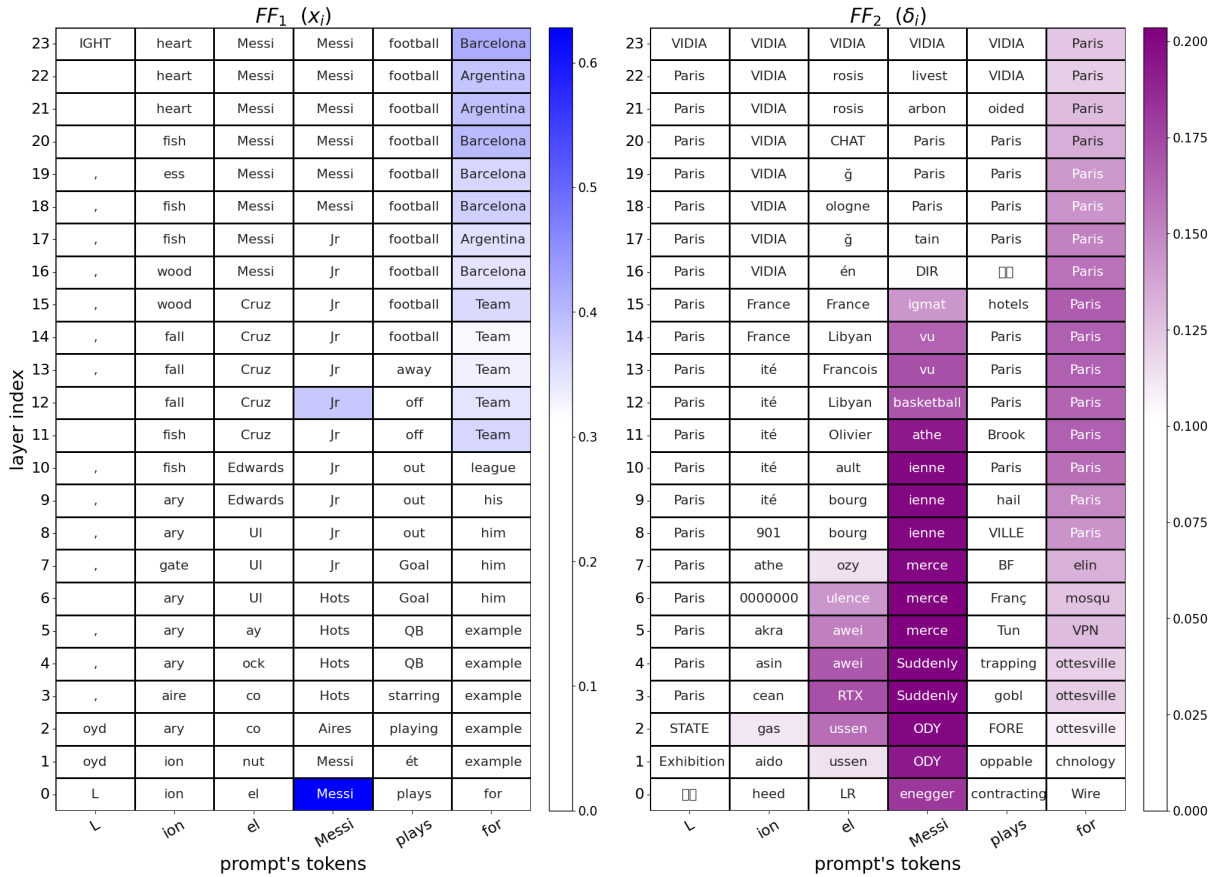


Figure 11: GPT2-medium MLP gradients via our spanning set interpretation. Each cell illustrates the Logit Lens of the gradient’s spanning set according to a layer and a token from the edited prompt. According to [Section 5.2](#) observation, for FF_1 we present the projections of the most probable tokens for the intermediate inputs x_i . For FF_2 we show most improbable tokens for the VJPs δ_i . The color indicates the norm of the gradient’s δ_i . In this example, the prompt is “Lionel Messi plays for”, to which the model responds with “Barcelona”. The new target is “Paris”. Notably, the target token “Paris” is evident through the majority of FF_2 ’s VJPs, which are the vectors that are injected into FF_2 ’s weights. On the other hand, FF_1 reflects not only the tokens that the gradients try to inject into FF_1 , but also the model’s intermediate predictions at each MLP layer during the forward pass. It is worth noting that the presence of tokens with a less clear meaning, such as “VIDIA” in FF_2 results from the projection of vectors with almost zero norm, as exemplified in (Table 2). This implies that these subcomponents of the gradient exert negligible influence when updating the model.

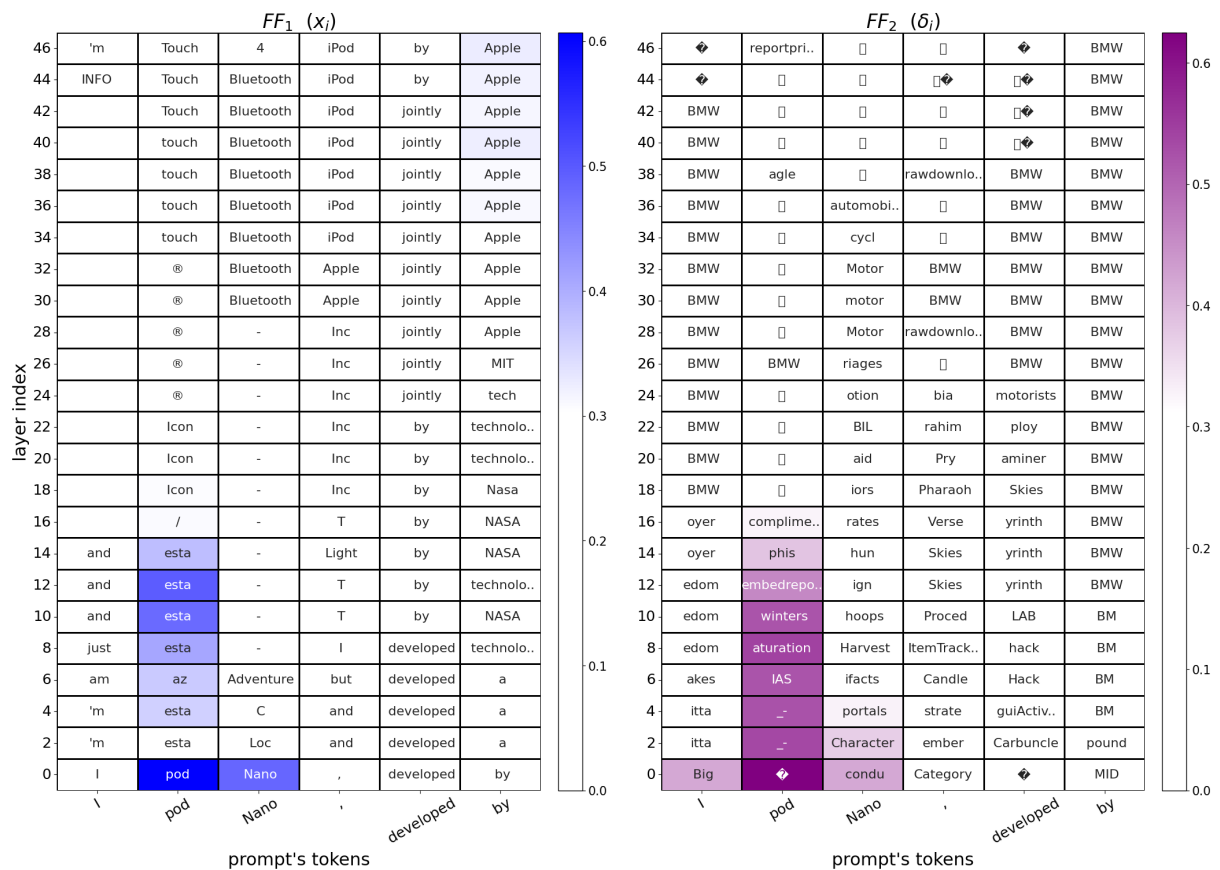


Figure 12: GPT2-xl MLP gradients via Logit Lens for “Ipod Nano, developed by” which we edit from “Apple” to “BMW”. Due lack of space, we show only every second layer. The color scheme indicates that the primary focus of editing lies on the subject token “Pod”. We hypothesize that the word “Ipod” is highly related to “Apple”, so by targeting “pod”, the gradients edit the relation between Ipod and the company that created it. Question marks and empty boxes are non-English tokens.